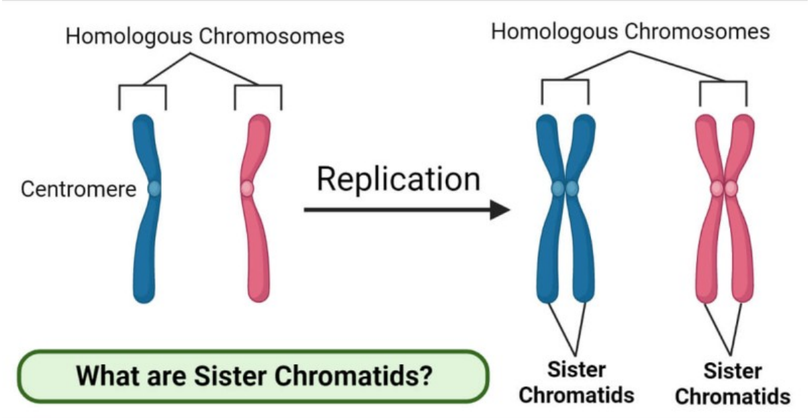
S C E D e t e c t o r

from MosaiCatcher Output Files

Hyunjin Cho



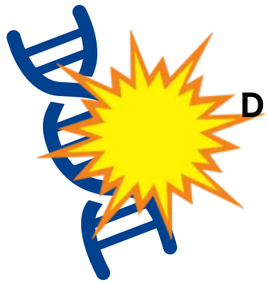
Sister Chromatid Exchange (SCE)



Exchange



why?



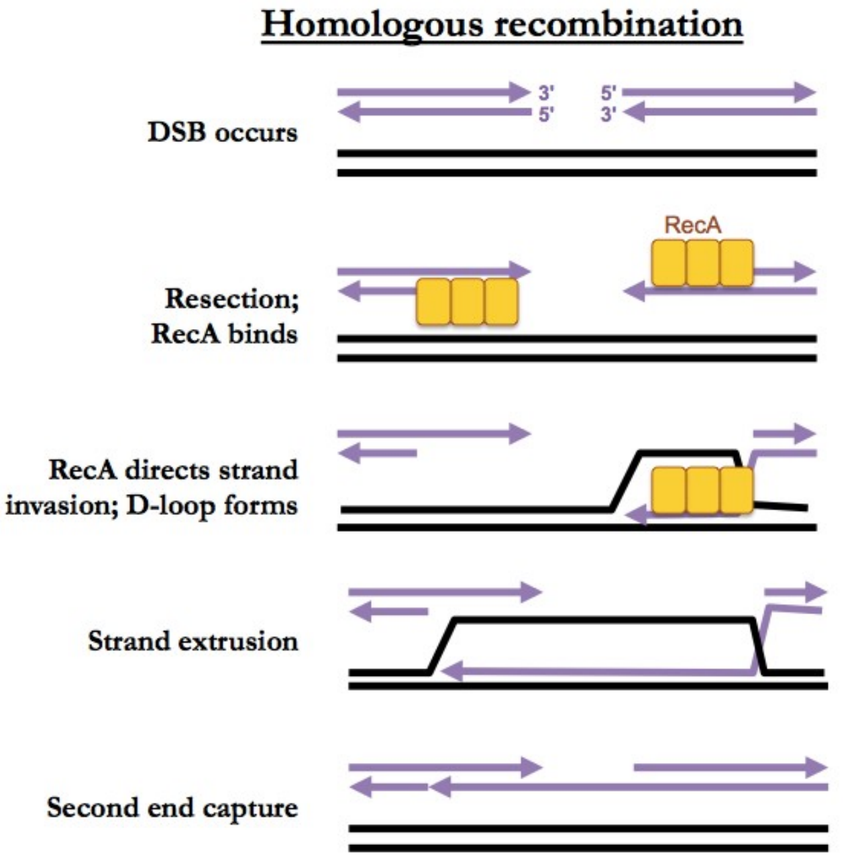
Damage

Homologous Recombination

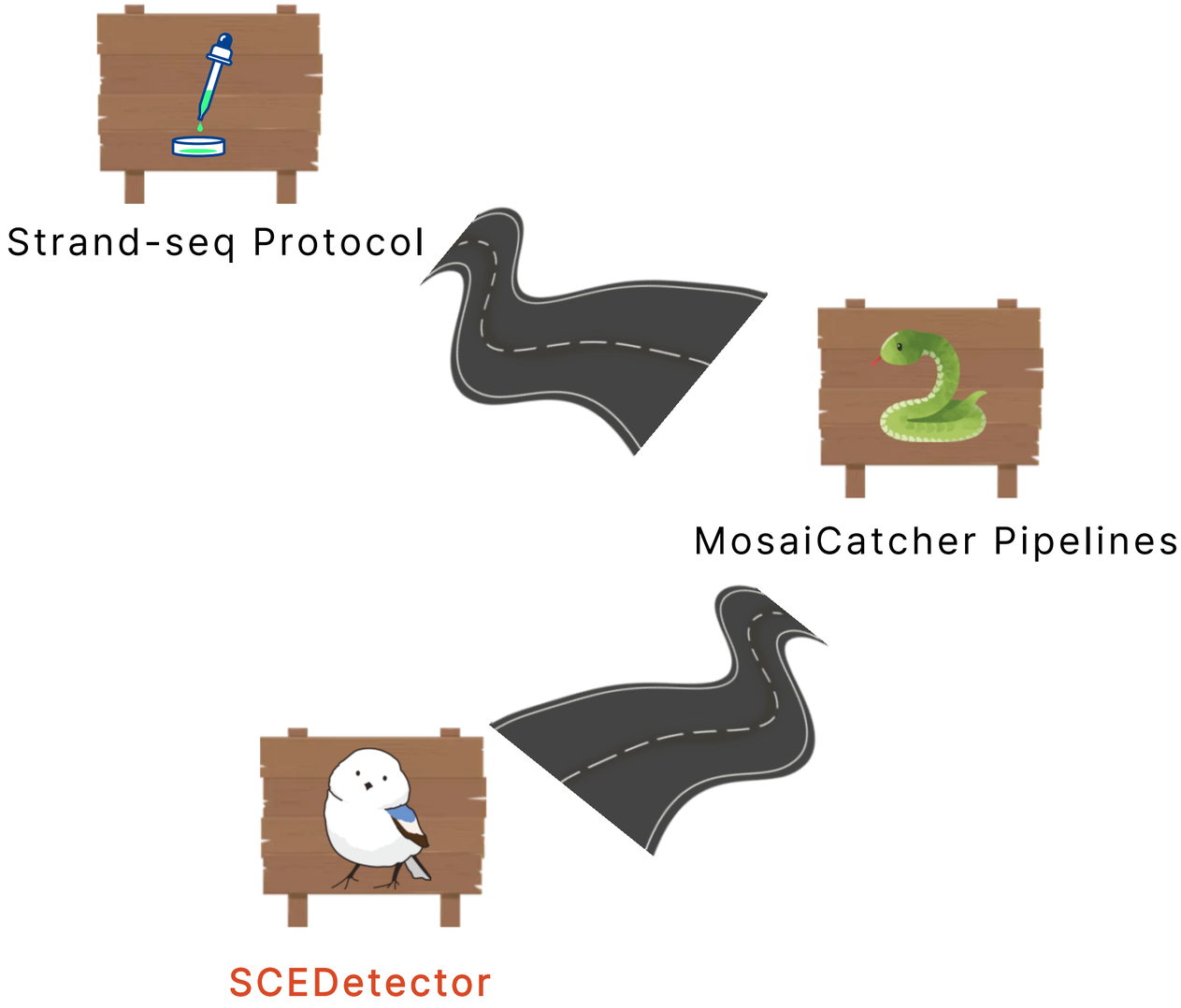
SCE

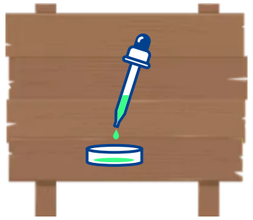
"Marker for Genomic Instability"

*

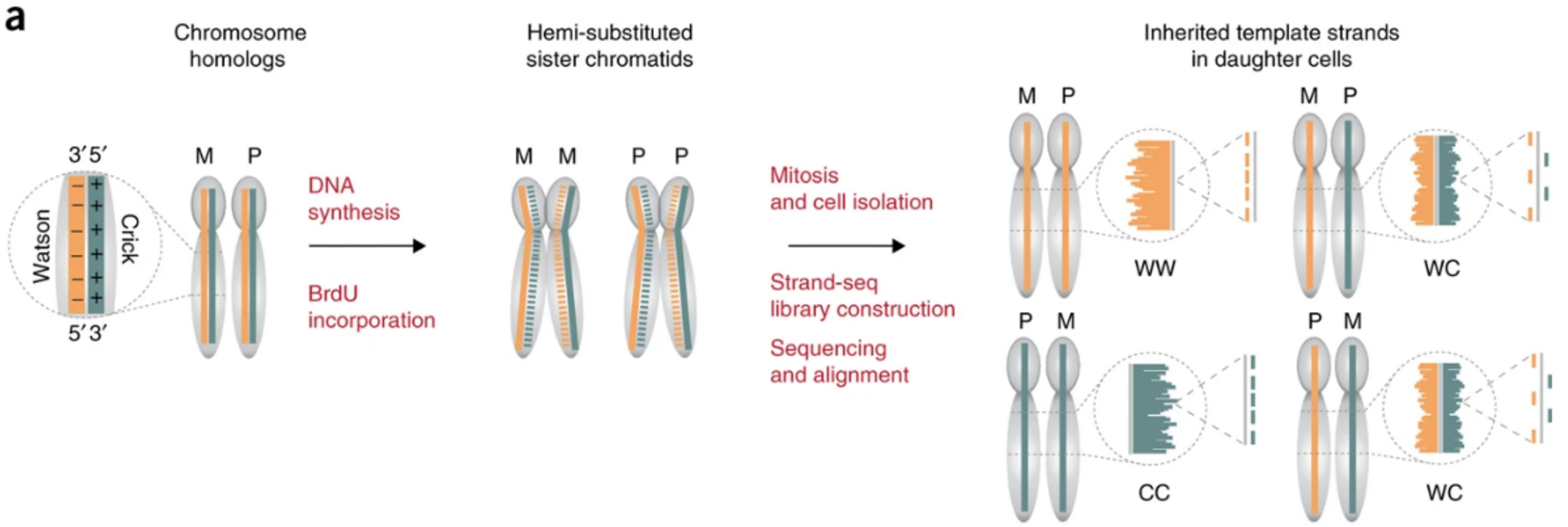


How to detect breakpoints of SCE





Strand-seq Protocol



Sanders et al., Nature Protocols

"Single-cell template strand sequencing by Strand-seq enables the characterization of individual homologs"



Saet Byeol Lee

Yujung Ahn

Surin Kim



MosaiCatcher Pipelines

MOSAIC CATCHER

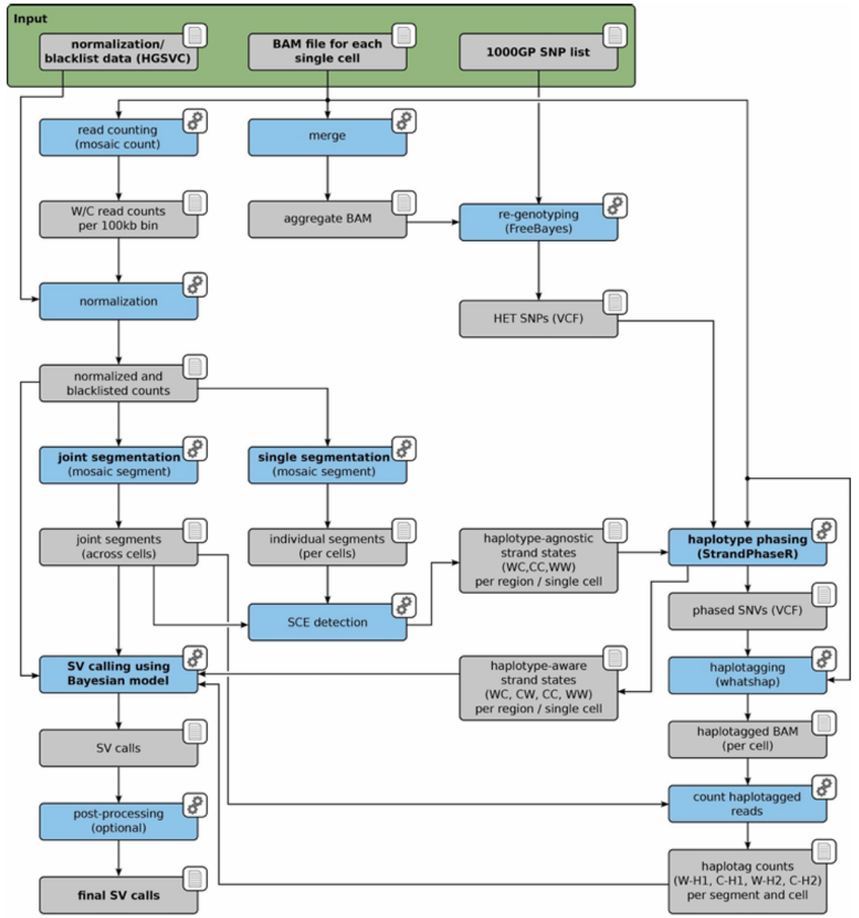
MosaiCatcher basic checks passing | snakemake >9.0 | License MIT | docs online | DOI 10.1093/bioinformatics/btad633 | release v2.5.0

ASHLEYS-QC-PIPELINE

ashleys-qc-pipeline workflow checks no status | snakemake ≥7.12.0

scNOVA

scNOVA : Single-Cell Nucleosome Occupancy and genetic Variation Analysis - summarized in a single Snakemake pipeline.



Sanders et al., Nature Biotechnology

"Single-cell analysis of structural variations and complex rearrangements with tri-channel processing"

Chiwon Chung



MosaiCatcher Pipelines

Output

Various Files

.....

lenient_filterFALSE.tsv

chromosome plot

StrandPhaseR_final_output.txt

(sample).txt.raw.gz

.....

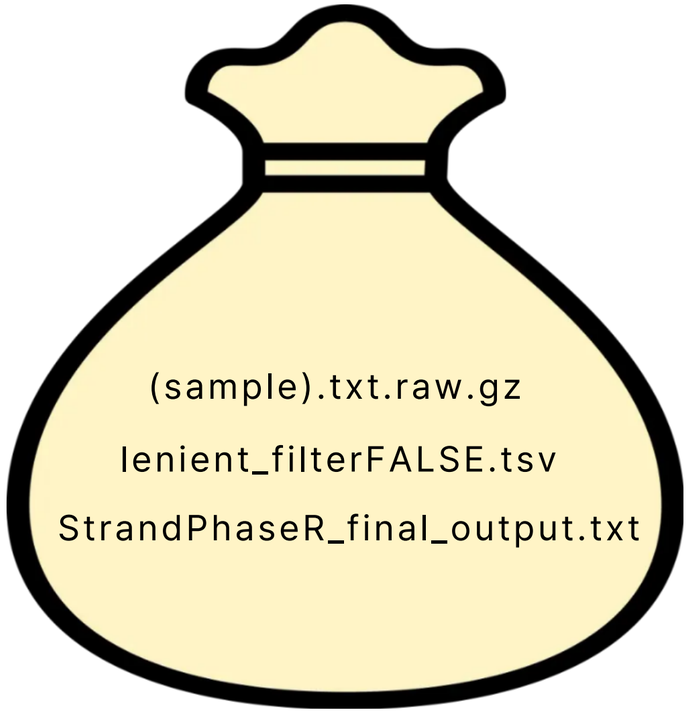


SCEDetector

Contributor

Hyunjin Cho: Developer
Prof. Hyobin Jeong: Advisor

Input

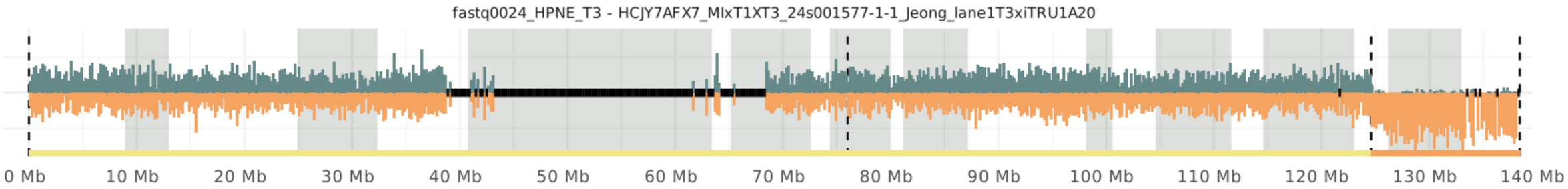


SCE_detected.xlsx



Sample	Cell_ID	chr	start
HPNE_M	~~G86	chr14	60200000

Quiz

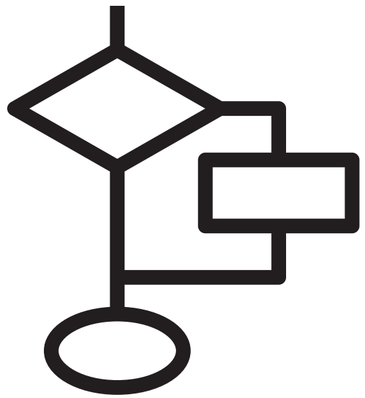




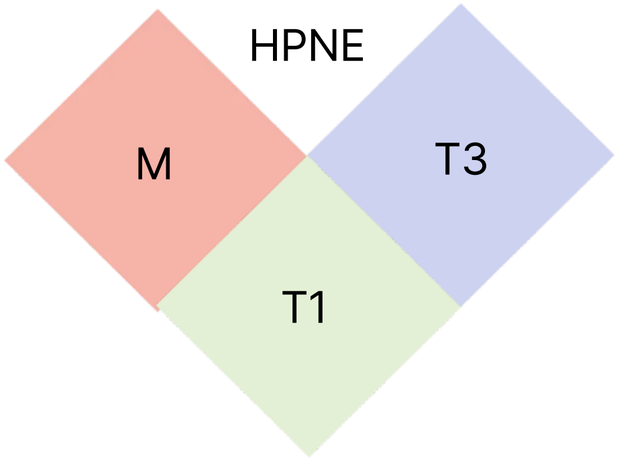
"Single State Change"

CC -> WC
WC -> WW
WC -> CC
WW -> WC

Heuristic Algorithm

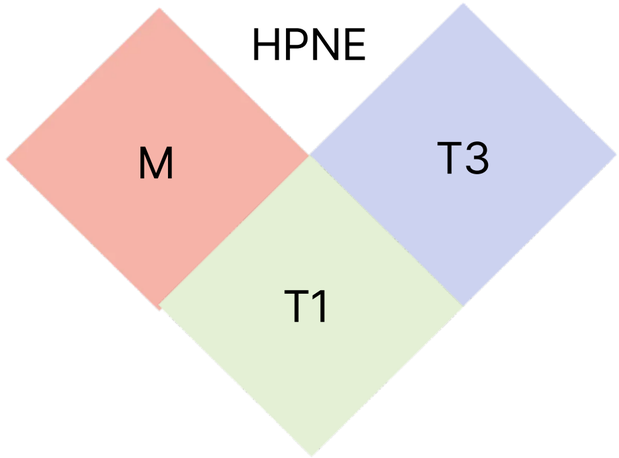


"A heuristic algorithm is a rule-based approach that finds good approximate solutions efficiently without guaranteeing the optimal solution"



of chromosomes: 736

of QC-passed cells: 32



of chromosomes: 782

of QC-passed cells: 34

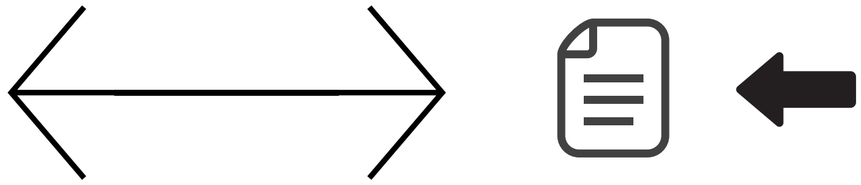
of chromosomes: 736

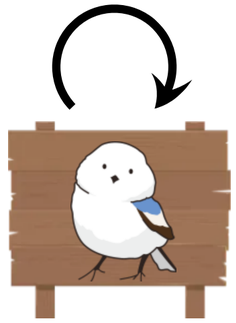
of QC-passed cells: 32

Update Heuristic Rules

Manual labeling of SCE

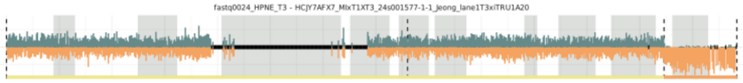
SCE_detected.xlsx





produced by Prof. Hyobin Jeong

SCEDetector



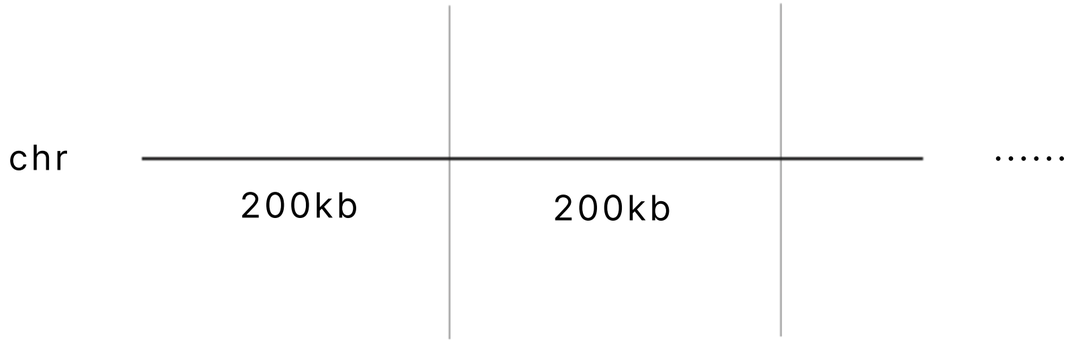
Chr Plot for Validation

"200kb bin identification"



(sample).txt.raw.gz

chr17	0	200000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	6	0	CC
chr17	200000	400000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	23	0	CC
chr17	400000	600000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	14	0	CC
chr17	600000	800000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	17	0	CC
chr17	800000	1000000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	19	0	CC
chr17	1000000	1200000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	22	0	CC
chr17	1200000	1400000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	15	0	CC
chr17	1400000	1600000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	14	0	CC
chr17	1600000	1800000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	23	0	CC
chr17	1800000	2000000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	12	0	CC
chr17	2000000	2200000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	14	0	CC
chr17	2200000	2400000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	15	0	CC
chr17	2400000	2600000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	11	0	CC
chr17	2600000	2800000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	15	0	CC
chr17	2800000	3000000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	17	0	CC
chr17	3000000	3200000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	13	0	CC
chr17	3200000	3400000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	11	0	CC
chr17	3400000	3600000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	20	0	CC
chr17	3600000	3800000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	13	0	CC
chr17	3800000	4000000	fastq0024_HPNE_T3	HCJY7AFX7_MIxT1XT3_24s001577-1-1_Jeong_lane1T3xiTRU1A05	16	0	CC



classified
by
Hidden Markov Model
(HMM)

involved in
MosaiCatcher

"Bin Masking"



HGSVC.200000.txt



chromosome_arm_positions_grch38.txt

Identification of low mappability or centromere regions for humans



We will update it for multiple species



Bin Masking Rule 1

mask: ... None | None | None ...
Usage: ... × | × | × ...

Bin Masking Rule 2

mask: None None | good good good | None None
↑ ≤5 bins ↑
Processed: None None | None None None | None None

Bin Masking Rule 3

mask: good good good good good | None×10bins+ | good good good ...
←—— max 5 bins ——→ ←—— max 5 bins ——→
Processed: ... None None None None | None... | None None ...

Bin Masking Rule 4

Remove centromere

[p.End, q.Start)

"Bin Masking"



Ienient_filterFALSE.tsv

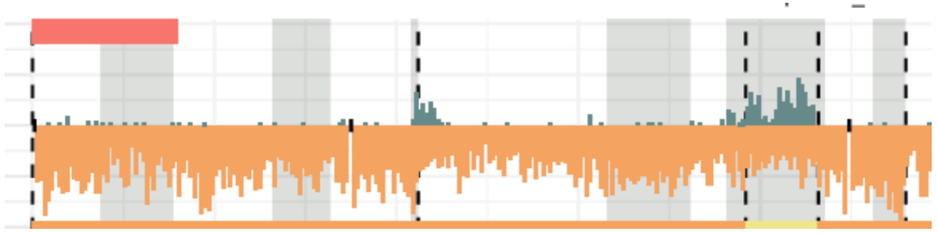
Identification of SV regions

Bin Masking Rule 5

Remove SV bins

SV	Processed
del, dup, idup	drop
inv, complex	drop (but drop x if it is on tips)

"Artifact Removal"



There are some cases that we should remove

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

balance $= \frac{\min(c, w)}{\max(c, w)}$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

Example: Calculation for "dom"

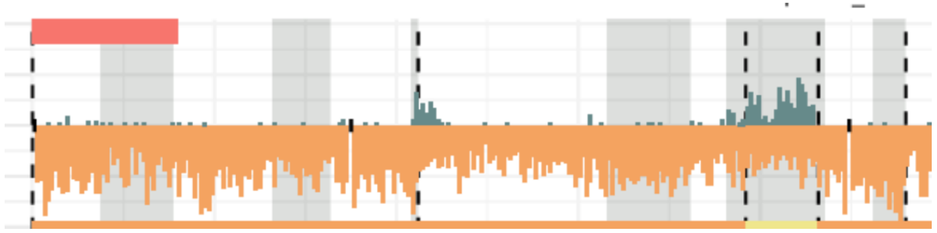
left WW (w median = 18) | WC | Right WW (w median = 22)

left_dom = 18 right_dom = 22

flank_mean = (18 + 22) / 2 = 20

dom = max(20, ref_depth)

"Artifact Removal"



There are some cases that we should remove

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

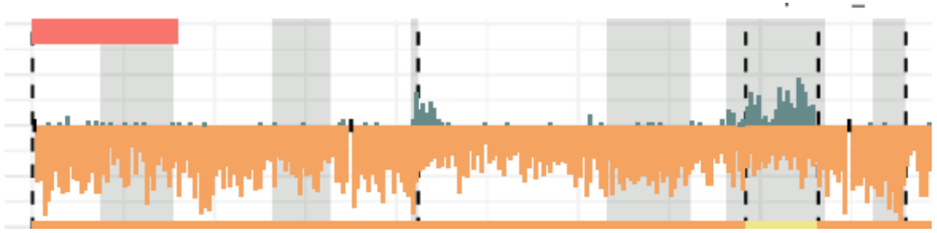
$$\text{balance} = \frac{\min(c, w)}{\max(c, w)}$$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

Example: "balance"

	C	W	balance
WC run 1	10	10	1
WC run 2	12	10	0.83
....	20	5	0.25
	20	0	0

"Artifact Removal"



There are some cases that we should remove

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

balance $= \frac{\min(c, w)}{\max(c, w)}$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

Expected coverage

Cell i's normal depth at bin b

E[depth(b, i) | cell i]

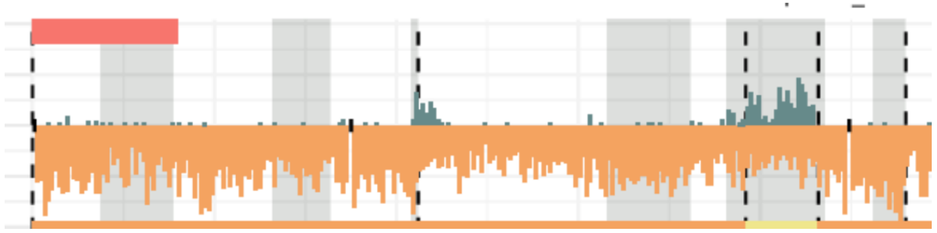
For a single chromosome,

"Assumption 1" → S_i (Cell Effect)
: If the total depth in a cell is high/low, the depth of all bins in the cell is proportionally high/low

"Assumption 2" → r_b (Bin Effect)
: There is the effect on the depth, which is only from the bin (not cell)

E[depth(b, i) | cell i] $\fallingdotseq$ $r_b \cdot s_i$

"Artifact Removal"



There are some cases that we should remove

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

balance $= \frac{\min(c, w)}{\max(c, w)}$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

Expected coverage

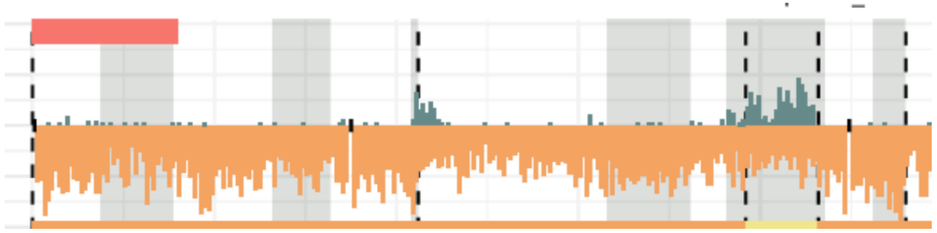
"Assumption 1" → S_i

: If the total depth in a cell is high/low, the depth of all bins in the cell is proportionally high/low

$$s_i = \text{median}_b \left(c_{b,i} + w_{b,i} \right) = \texttt{scale}[i]$$

median depth of cell i

"Artifact Removal"



There are some cases that we should remove

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

balance $= \frac{\min(c, w)}{\max(c, w)}$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

Expected coverage

"Assumption 2" → r_b
: There is the effect on the depth, which is only from the bin (not cell)

$$d_{b,i} = c_{b,i} + w_{b,i}$$

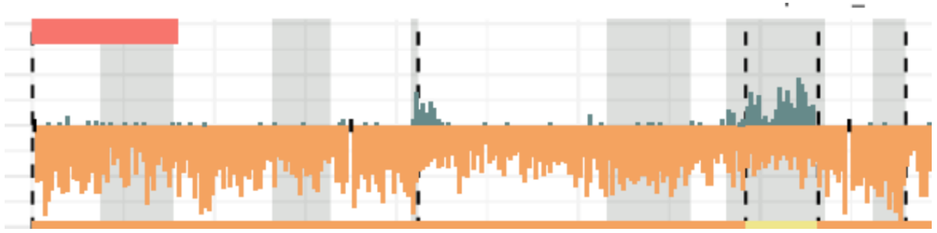
$$r_b \doteqdot \frac{\text{E[depth(b, i) | cell i]}}{s_i} \doteqdot \text{median}_{i \in \text{QC cells}} \left(\frac{d_{b,i}}{s_i} \right) = \texttt{bin_rel}[b]$$

median across all cells

→ Excluding the effects or noises from cells

→ "Assumption 3"
: More than 50% of observed cells have normal depth

"Artifact Removal"



There are some cases that we should remove

Expected coverage

Variables for Rules

C: median depth of C

W: median depth of W

dom: median depth of flanking homozygous run(s)

balance $= \frac{\min(c, w)}{\max(c, w)}$ only in WC runs

ref_depth: median depth of all homozygous runs in a cell

E[depth(b, i) | cell i] $\fallingdotseq$ $r_b \cdot s_i = \texttt{bin_rel}[b] \times \texttt{scale}[i]$

Cell i's normal depth at bin b

"Artifact Removal"

Artifact Removal Rule 1

Drop duplication-like WC

MosaiCatcher sometimes does not capture duplication-like WC.... 🥺

Condition

$$\frac{\max(c, w)}{\text{dom}} \geq 0.85$$

AND

$$\frac{\min(c, w)}{\text{dom}} \geq 0.35$$

AND

$$\frac{c + w}{\text{dom}} \geq 1.35$$

OR

$$(c + w) > 1.50 \cdot \text{expected}$$

expected coverage

⬇

Duplication-like WC

"Artifact Removal"

Artifact Removal Rule 2

Drop short stubs at SV hole edges

The regions range called as SV by MosaiCatcher is sometimes narrower than the real.... 🥺

WW WW WW | [SV hole] | WC WC | WW WW WW
↑ stub (≤2 bins)

400kb

$$\text{drop run if } (\#\text{bins} \leq 2) \ \wedge \ (\text{run touches SV skip})$$

kept WC 410kb |======== SV hole 10 Mb ========| kept WC 410kb

(1) run touches SV skip

(2) $\text{kept} \leq \begin{cases} 2.0\,\text{Mb} & \text{if class} = \text{WC} \\ 0.8\,\text{Mb} & \text{if class} = \text{WW or CC} \end{cases}$

empirically WC is easy to make noisy stubs (Heuristic)

(3) hole > kept

"Artifact Removal"

Artifact Removal Rule 3

Drop WC island in A-WC-A patterns

e.g. WW - WC - WW

CC - WC - CC

Positive: Inversion or Double SCE

Negative: Noisy WC island

In WW-WC-WW, WC run is actually WW, but it is misclassified as WC because of noise C

Condition

$$\text{balance} > 0.75$$

OR

$$L \geq 10\,\text{Mb} \;\wedge\; \text{balance} \geq 0.50$$

Length of WC

OR

$$(c + w) < 0.55 \cdot \text{dom}$$

OR

$$\text{balance} \geq 0.35 \;\wedge\; \text{island_dom} \leq 0.47 \cdot \max(\text{flank_doms})$$

if flank_doms are Watson strand, island_dom is also W strand.

OR

$$\begin{aligned} &L \leq 5\,\text{Mb}, \quad \text{balance} \geq 0.55, \\ &\min(\text{flank_doms}) \leq (c + w) < \max(\text{flank_doms}), \\ &\text{island_dom} > 0.55 \cdot \max(\text{flank_doms}), \\ &\min(c, w) \geq 0.45 \cdot \min(\text{flank_doms}) \end{aligned}$$

min flank WW (dom≈12) | WC island WC (tot≈16) | max flank WW (dom≈22)

←L ≤ 5 Mb→

WC run is kept → Next filter

otherwise

drop

"Artifact Removal"

Artifact Removal Rule 3

Drop WC island in A-WC-A patterns

e.g. WW - WC - WW

CC - WC - CC

→ Positive: Inversion or Double SCE

Negative: Noisy WC island

left flank gap WC island gap right flank gap

CC ..... |?????????| WC WC ... WC |????????| CC

Condition

WC length <= 1 Mb

AND

At least one flank gap >= 10 Mb

AND

uncovered <= 1 Mb

→ Drop

- uncovered

(SV cannot sometimes cover all gap regions)

"Artifact Removal"

Artifact Removal Rule 4

Drop WW/CC island in WC-A-WC patterns

e.g. WC - WW - WC

WC - CC - WC



Positive: Inversion or Double SCE

Negative: short artifact

or deletion-like WW/CC

Condition

AND

$$L < 5\,\mathrm{Mb}$$

$$(c+w)_{\text{island}} < 0.80 \times (c+w)_{\text{deeper WC}}$$

$$(c+w)_{\text{island}} < (c+w)_{\text{shallower WC}}$$



Short artifact WW/CC

→ Drop

"Artifact Removal"

Artifact Removal Rule 4

Drop WW/CC island in WC-A-WC patterns

e.g. WC - WW - WC

WC - CC - WC

→ Positive: Inversion or Double SCE

Negative: short artifact

or deletion-like WW/CC

$$\text{del cover} = \frac{\text{length of island overlapping } del_ * \text{ calls}}{\text{island length}}$$

Case that the region includes deletion regions called by MosaiCatcher

Condition

OR

$$\text{del cover} \geq 0.20 \ \wedge \ (c+w)_{\text{island}} < 0.80 \times (c+w)_{\text{deeper WC}}$$

$$\text{del cover} \geq 0.15 \ \wedge \ (c+w)_{\text{island}} < 0.40 \times (c+w)_{\text{deeper WC}}$$

→ deletion-like WW/CC

→ Drop

"Artifact Removal"

Artifact Removal Rule 4

Drop WW/CC island in WC-A-WC patterns

e.g. WC - WW - WC

WC - CC - WC



Positive: Inversion or Double SCE

Negative: short artifact

or deletion-like WW/CC

It can cover the case that the region does not include deletion regions called by MosaiCatcher

Condition

AND

$L \geq 5\,\mathrm{Mb}$

$\mathrm{del_cover} < 0.15$

$(c+w)_{\mathrm{island}} < 0.40 \times (c+w)_{\mathrm{deeper\ WC}}$

$(c+w)_{\mathrm{island}} < (c+w)_{\mathrm{shallower\ WC}}$



deletion-like WW/CC

→ Drop

"Artifact Removal"

Artifact Removal Rule 5

Drop WW/CC tips from WC-WW/CC-end patterns

e.g. WC - WW

WC - CC



Positive: SCE

Negative: short artifact

or deletion-like WW/CC

Condition

AND

$$\text{tip abuts WC (no masked gap)}$$

$$(c+w)_{\text{tip}} < 0.70 \times (c+w)_{\text{WC}}$$

$$\text{del_cover(tip)} \geq 0.10$$

$$\exists\, del \text{ within } 2\,\text{Mb of the WC} \leftrightarrow \text{homo junction}$$



Drop

"Artifact Removal"

Artifact Removal Rule 5

Drop WW/CC tips from WC-WW/CC-end patterns

e.g. WC - WW

WC - CC

➡

Positive: SCE

Negative: short artifact

or deletion-like WW/CC

Chr has only 2 runs

Condition

AND

$$\text{tip abuts WC (no masked gap)}$$

$$(c+w)_{\text{tip}} < 0.70 \times (c+w)_{\text{WC}}$$

$$\text{del_cover(tip)} \geq 0.10$$

$$\exists\, del \text{ within } 2\,\text{Mb of the WC} \leftrightarrow \text{homo junction}$$

➡ Drop

"Artifact Removal"

Artifact Removal Rule 5

Drop WW/CC tips from WC-WW/CC-end patterns

e.g. WC - WW

WC - CC



Positive: SCE

Negative: short artifact

or deletion-like WW/CC

Chr has any runs

Condition

AND

$$L_{\text{tip}} \leq 15\,\text{Mb}$$

$$(c+w)_{\text{tip}} \leq 0.70 \times (c+w)_{\text{WC}}$$



Drop

"Artifact Removal"

Artifact Removal Rule 6

Acrocentric chr [masked p/cen] | WW/CC stub (≤1 Mb) | WC ---------- q-arm

Drop

$$\text{chrom} \in \{13, 14, 15, 21, 22\}$$
$$\wedge \text{ first kept run is } WW/CC$$
$$\wedge \; L_{\text{stub}} \leq 1\,\text{Mb}$$
$$\wedge \text{ stub abuts following WC}$$

➡ Drop

"Double-Switch Extraction and Resolution"

Double-Switch Rule 1

A-B-A Sandwich

e.g.

WW – WC – WW
CC – WC – CC

Priority	Condition	Result
1	Both breakpoints shared within ±10 kb by ≥5% of QC cells	Inversion
2	Exactly one inversion SV call lies inside the sandwich and covers ≥ 40% of it	Inversion
3	Otherwise	Two SCE

Double-Switch Rule 2

Two-step opposite

e.g.

WW – WC – CC
CC – WC – WW

➡ Two SCE

"Single Switch SCE"

Single-Switch Rule 1



Check one single state change in each arm

Check one single state change in each arm

Single-Switch Rule 2

if there are many and ambiguous breakpoints, we choose the right start point as a breakpoint



if a single breakpoint (WW → WC) is in the masked area, we choose it as the breakpoint

"Breakpoint refinement across SV holes"

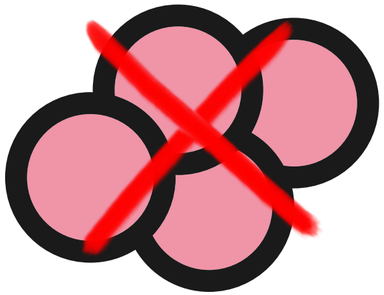


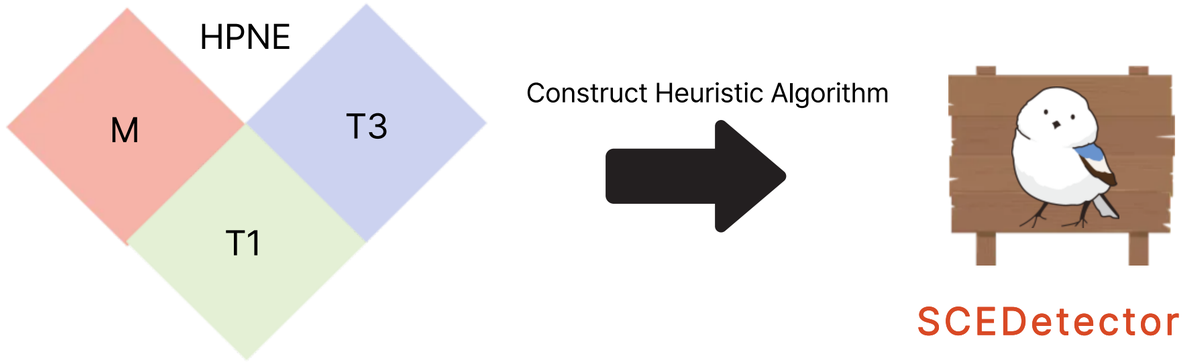
"Recurrence -> Translocation"

Breakpoints shared within ±10 kb by ≥5% of QC cells ➡ Translocation



SCE do not form the clones or subclones





How to validate SCEDetector?

Stop updating algorithm

→ Validation through other Strand-seq samples?



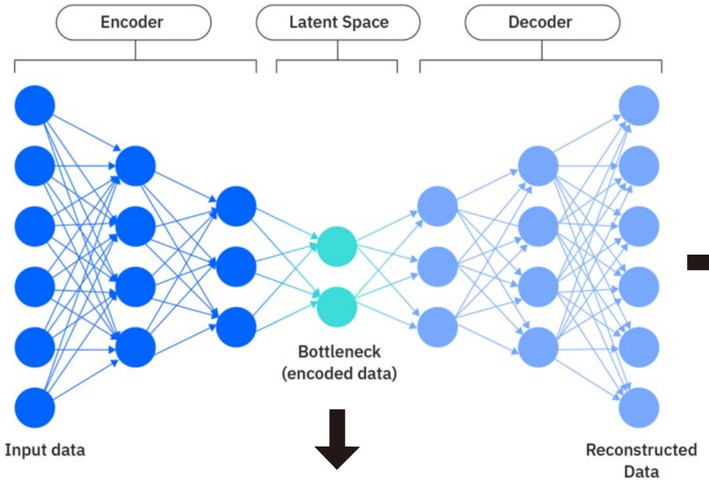
Application?

SCEDetector

Input vector with some features from Strand-seq

1. SV breakpoints from MosaiCatcher
2. Nucleosome occupancy from scNOVA
3. SCE breakpoints from SCEDetector

StrandVI



Augmented Strand-seq Data

Dimension Reduction & Embedding (UMAP)